

Enemy of the State: A State-Aware Black-Box Web Vulnerability Scanner

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Black-box web vulnerability scanners are a popular choice for finding security vulnerabilities in web applications in an automated fashion. These tools operate in a **point-and-shoot** manner, testing any web application—regardless of the server-side language—for common security vulnerabilities. Unfortunately, black-box tools suffer from a number of limitations, particularly when interacting with complex applications that have multiple actions that can change the application’s state. If a vulnerability analysis tool does not take into account changes in the web application’s state, it might overlook vulnerabilities or completely miss entire portions of the web application.

We propose a novel way of inferring the web application’s internal state machine *from the outside*—that is, by navigating through the web application, observing differences in output, and incrementally producing a model representing the web application’s state.

We utilize the inferred state machine to drive a black-box web application vulnerability scanner. Our scanner traverses a web application’s state machine to find and fuzz user-input vectors and

discover security flaws. We implemented our technique in a prototype crawler and linked it to the fuzzing component from an open-source web vulnerability scanner.

We show that our state-aware black-box web vulnerability scanner is able to not only exercise more code of the web application, but also discover vulnerabilities that other vulnerability scanners miss.

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